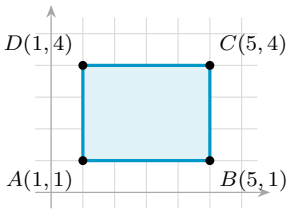
Perimeter & Area in the Coordinate Plane — Notes — Key

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: Side lengths from coordinates*

Key Idea



Rectangle $ABCD$:

$AB = 5 - 1 =$ 4 $\qquad BC = 4 - 1 =$ 3

Same y: subtract the x's. Same x: subtract the y's.

☞ *Video: Perimeter and area*

Key Idea

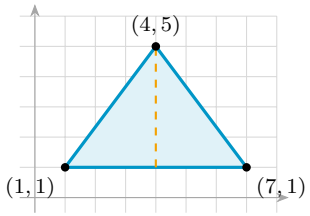
Perimeter = add all the sides : $P = 4 + 3 + 4 + 3 =$ 14

Area = base $\times$ height : $A = 4 \cdot 3 =$ 12

Perimeter is a length: units . Area counts squares: square units .

☞ *Video: Area of a triangle*

Key Idea



base $= 7 - 1 =$ 6

height $= 5 - 1 =$ 4 (straight up from the base)

$A =$ $\frac{1}{2}$ $\cdot b \cdot h = \frac{1}{2} \cdot 6 \cdot 4 =$ 12

☞ *Video: A slanted side? Use Pythagoras*

Key Idea

Same triangle. A slanted side goes across 3 and up 4: $\sqrt{3^2 + 4^2} = \sqrt{25} =$ 5 .

The other slanted side is also 5. $P = 6 + 5 + 5 =$ 16

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☞ *Video: Watch out*

Common Mistake

For a **slanted** side you can't just count squares. Use the **Distance Formula** — or draw the right triangle and use **Pythagoras**.

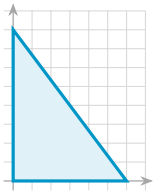
☞ *Video: Your turn*

Your Turn

Triangle $P(0,0)$, $Q(6,0)$, $R(0,8)$. Find its perimeter and area.

$PQ = 6,\ PR = 8, \quad QR = \sqrt{6^2 + 8^2} =$ 10

Perimeter $= 6 + 8 + 10 =$ 24 Area $= \frac{1}{2} \cdot 6 \cdot 8 =$ 24



Notice

Remember: Same y: subtract the x's; same x: subtract the y's. Perimeter: add the sides. Area: rectangle $b \cdot h$, triangle $\frac{1}{2}bh$. Slanted side: Pythagoras or the Distance Formula.

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